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SAAZA Process

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Purpose

- The purpose of this presentation is to prepare prospective surveyors to properly execute all four phases of the SAAZA process.



Learning Objectives

- Identify the four phases of a survey
- Identify key information and resources used during the information gathering phase
- Identify planning considerations during the preparation phase
- Identify priorities during the site visit
- Identify the forms required to conduct final product development



References

- MMT TACSOP
- UFC 3-260-1 Airfield and Heliport Planning and Design
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
- DAFMAN 13-217 Nuclear, Space, Missile, Command and Control; Drop Zone Landing Zone and Helicopter Landing Zone Procedures



References

- Engineering Technical Letter (ETL) 09-6 (17 Aug 09)
- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures



Outline

- Introduction / Overview
- Unit One: Mathematics and Measurement
- Unit Two: Landing Zone Criteria
- Unit Three: Soil Identification and Evaluation
- **Unit Four: SAAZA Process**
- Unit Five: SAAZA Completion



Outline

- Overview
- Information Gathering
- Preparation
- Site Visit
- Final Product Development
- Review
- Summary



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Overview

- Survey is a process
 - Planning and preparation is essential
 - Physical survey is the simplest portion
 - Everything is invisible except the final product
- The Seven “P” Rule



Overview

- Four phases
 - Information Gathering
 - Collection and consolidation of all pre-existing information on a particular location
 - Preparation
 - Team planning, task assignment, logistics
 - Site Visit
 - Physical collection of ground truth data
 - Final Product Development
 - Synthesis of all data points into cohesive and informative document for requesting agency

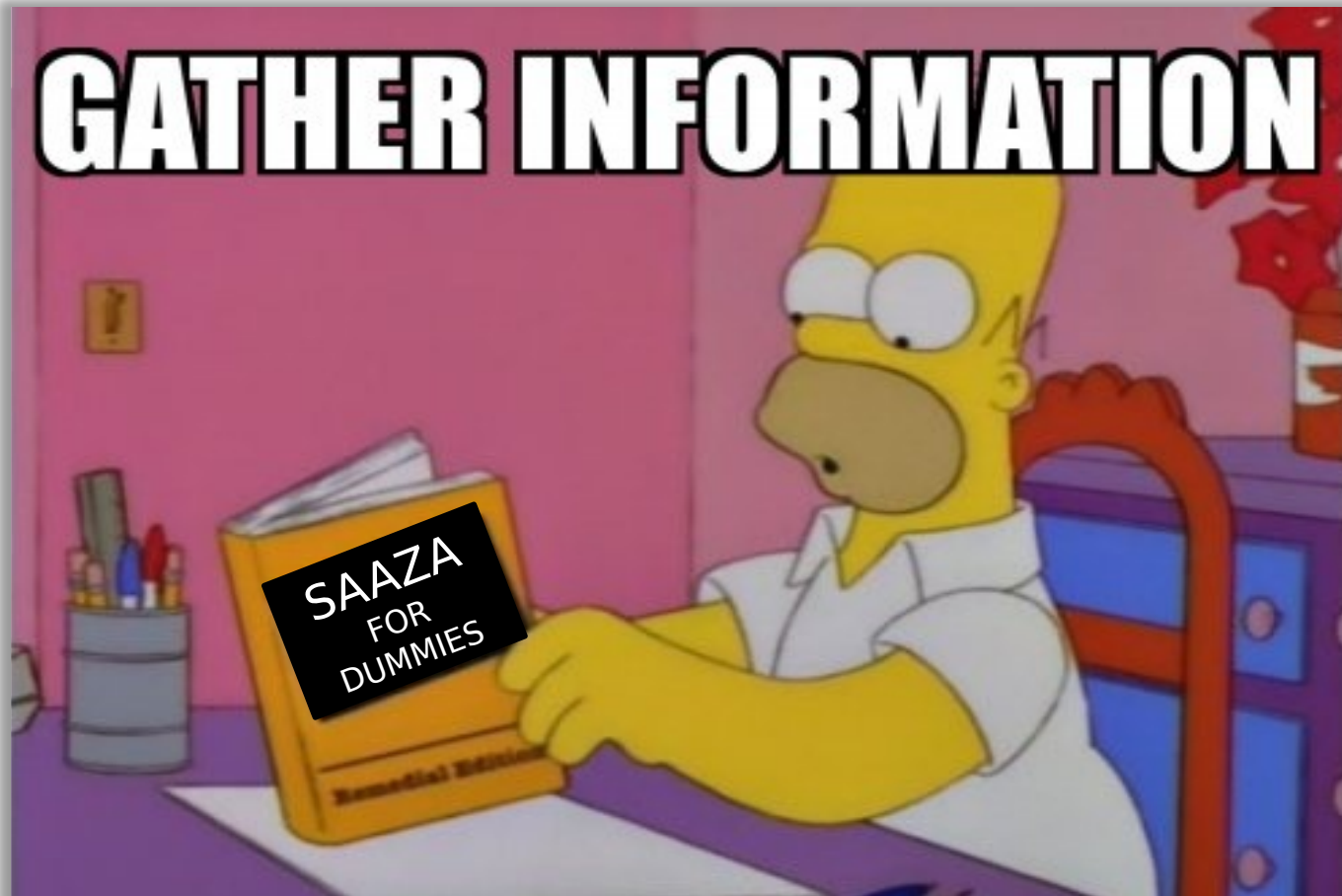


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Information Gathering





Information Gathering

- Existing information assists in creating an efficient and effective collection plan for the Site Visit
 - Creates priorities to fill in gaps in information
 - Information must still be validated
 - Provides a good indicator of the amount of collection required to accomplish the mission



Information Gathering

- Previously completed surveys
- Satellite imagery
- Maps
- ATC supplemental info
- Weather reports
- Intelligence reports
- Host Nation (HN) points of contact for controlling agencies
- Requestor criteria
- Available assets



Information Gathering

- Previously completed surveys
 - Best source of info for a specific location
 - May still be current
 - Will outline previous areas of concern
- Talon Point and the Zone Availability Reports (ZARs)
 - NIPR

<https://talonpoint.net/>

- SIPR

<https://talonpoint.snlca.nro.smil.mil/TalonPoint/home>



Information Gathering

- Satellite Imagery
 - S-2 Products
 - Google Earth
 - HAWG-VIEW
- Maps
 - Airspace
 - Range
 - Road / relief



Information Gathering

- Intelligence reports
 - Security concerns
- ATC supplemental information
 - Factors that affect ATC operations
 - National Geospatial Agency (NGA)
 - Weather reports
 - Host Nation (HN) points of contact for controlling agencies



Information Gathering

- Requestor criteria
 - Ensure you have all T/M/S specific criteria information found in NATOPS Manuals
 - Glideslopes
 - Gradients
 - Airframe mission weight for fixed-wing
 - LZ size requirements
 - Supplemental requests
 - Your assessment depends on this information



Information Gathering

- Available assets
 - Logistics
 - Air or Ground Insert
 - Required life support for time on station
 - Personnel
 - EAF integration
 - Security attachment
 - Equipment



Information Gathering

- Transitioning to preparation
 - Determine what information you have to determine what information you do not have
 - Prioritize missing information and make a list of necessary tasks to accomplish the mission



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Preparation

- A team leader prepares the team for mission success through detailed planning





Preparation

- Mission requirements
- Limitations
- Available assets
- Task organization
- Mission briefing
- Rehearsals
- Equipment
- Staging



Preparation

- Mission requirements
 - LZ purpose (ALZ, HLZ, FARP, etc)
 - Requesting agency (T/M/S)
 - Supplemental requests



Preparation

- Limitations
 - Time (including time of day)
 - Personnel
 - Security concerns



Preparation

- Available assets
 - Ground or Air insert
 - EAF personnel
 - Security personnel



Preparation

- Task organization
 - Specific tasking for every member
 - Primary, secondary, tertiary tasks
 - Integrate EAF / security personnel



Preparation

- Mission briefing
 - Team Leader must paint a clear and concise picture of the ground scheme of maneuver to every member and all integrated personnel
 - Especially important if there is a time constraint
 - Clear expectations of outputs



Preparation

- Rehearsals
 - Time permitting, Team Leaders should walk their team and integrated personnel through the scheme of maneuver
 - Builds confidence and understanding



Preparation

- Equipment
 - Function check all equipment
 - Pack out individual requirements



Preparation

- Staging
 - Mission has been planned, briefed, rehearsed, and equipped
 - Stage for embark and departure



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Site Visit





Site Visit

- This is the simplest phase of the process
 - Just execute your plan...
 - Collect data properly
 - ...and get out
- Phase where errors have the potential to create critical failures
 - Unlikely to get a second chance
 - Bad data = bad survey
 - Bad survey = loss of confidence



Site Visit

- Time
- Scope
- Manpower
- Equipment



Site Visit

- Time
 - Most critical limitation
 - Defines the scope of the survey
 - Hasty Survey and Assessment <6 hours
 - Full Survey and Assessment ≥ 12 hours



Site Visit

- Scope
 - Hasty
 - Will focus on presence of minimum criteria only
 - Specific to T/M/S
 - Full
 - Provide maximum ground truth data
 - Still specific to T/M/S
 - Provides enough information to tailor to other T/M/S if needed



Site Visit

- Manpower
 - Supervision of all members and integrated personnel
 - Ensure the plan is executed as briefed, or adjusted as necessary
 - Participation
 - The Team Leader is not solely a supervisor during survey and assessment



Site Visit

- Equipment
 - Accountability of gear at all times!
 - A DCP cone is an indicator, to the enemy



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Final Product Development

LANDING ZONE SURVEY	1A. LZ NAME	1B. ZAR INDEX NO.	2A. COUNTRY	2B. STATE
	3. MAP SERIES/SHEET NUMBER/EDITION/DATE OF MAP			
4. SURVEY APPROVAL/DISAPPROVAL DATA				
4A. DATE SURVEYED	TYPED NAME AND GRADE OF SURVEYOR	PHONE NUMBER (DSN)	UNIT	
4B. DATE REVIEWED	TYPED NAME AND GRADE OF REVIEWER	PHONE NUMBER (DSN)	SIGNATURE	
	UNIT AND LOCATION			
4C. DATE	TYPED NAME AND GRADE OF APPROVING AUTHORITY	PHONE NUMBER (DSN)	SIGNATURE	
APPROVED DISAPPROVED	UNIT AND LOCATION			

HELICOPTER LANDING ZONE SURVEY	1A. HLZ NAME	1B. ZAR INDEX NO.	2A. COUNTRY	2B. STATE
	3. MAP SERIES/SHEET NUMBER/EDITION/DATE OF MAP			
4. SURVEY APPROVAL/DISAPPROVAL DATA				
4A. DATE SURVEYED	TYPED NAME AND GRADE OF SURVEYOR	PHONE NUMBER (DSN)	UNIT	
4B. DATE REVIEWED	TYPED NAME AND GRADE OF REVIEWER	PHONE NUMBER (DSN)	SIGNATURE	
	UNIT AND LOCATION			
4C. DATE	TYPED NAME AND GRADE OF APPROVING AUTHORITY	PHONE NUMBER (DSN)	SIGNATURE	
APPROVED DISAPPROVED	UNIT AND LOCATION			



Final Product Development

- Data analysis and synthesis
- Product refinement
- Survey forms



Final Product Development

- Data analysis and synthesis
 - Compile information from all sources and condense into requirements
 - ALZ soil evaluation
 - 37 DCP test locations, 3 layers per test location
 - 111 CBR calculations
 - 1 controlling test location with 3 CBR calculations
 - 1 controlling layer with 1 CBR rating
 - 1 answer for number of allowable passes



Final Product Development

- Data analysis and synthesis
 - Gradients require calculations
 - ALZ gradients
 - 150 20' longitudinal gradient readings
 - 15 200' averages calculated
 - 1 3000' average calculated
 - 28 15' transverse gradient readings
 - 7 60' averages calculated



Final Product Development

- Data analysis and synthesis
 - HLZ gradients
 - Number of readings depends on HLZ size
 - Required to gather longitudinal and transverse



Final Product Development

- Data analysis and synthesis
 - Surfaces
 - Imaginary surface info
 - Obstacle data
 - Trigonometry



Final Product Development

- Product refinement
 - LZ diagram
 - Pictures
 - Survey forms



Final Product Development

- Survey Forms
 - Fixed-wing (ALZ): AF IRT 3822
 - Helicopter / Tilt-rotor (HLZ): AF IRT 4303



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Review

- Identify the four phases of a survey
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Review

- Identify key information and resources used during the information gathering phase
 - Talon Point
 - Satellite Imagery
 - Maps
 - Intel reports
 - HN support
 - T/M/S NATOPS
 - ATC supplemental info



Review

- Identify planning considerations during the preparation phase
 - Mission requirements
 - Limitations
 - Available assets
 - Task organization
 - Mission briefing
 - Rehearsals
 - Equipment
 - Staging



Review

- Identify priorities during the site visit
 - Time
 - Determines scope
 - Scope
 - Hasty vs. Full
 - Manpower
 - Supervision and execution
 - Equipment
 - Accountability



Review

- Identify the forms required to conduct final product development
 - Fixed-wing (ALZ): AF IRT 3822
 - Helicopter / Tilt-rotor (HLZ): AF IRT 4303



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Questions / Critiques



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